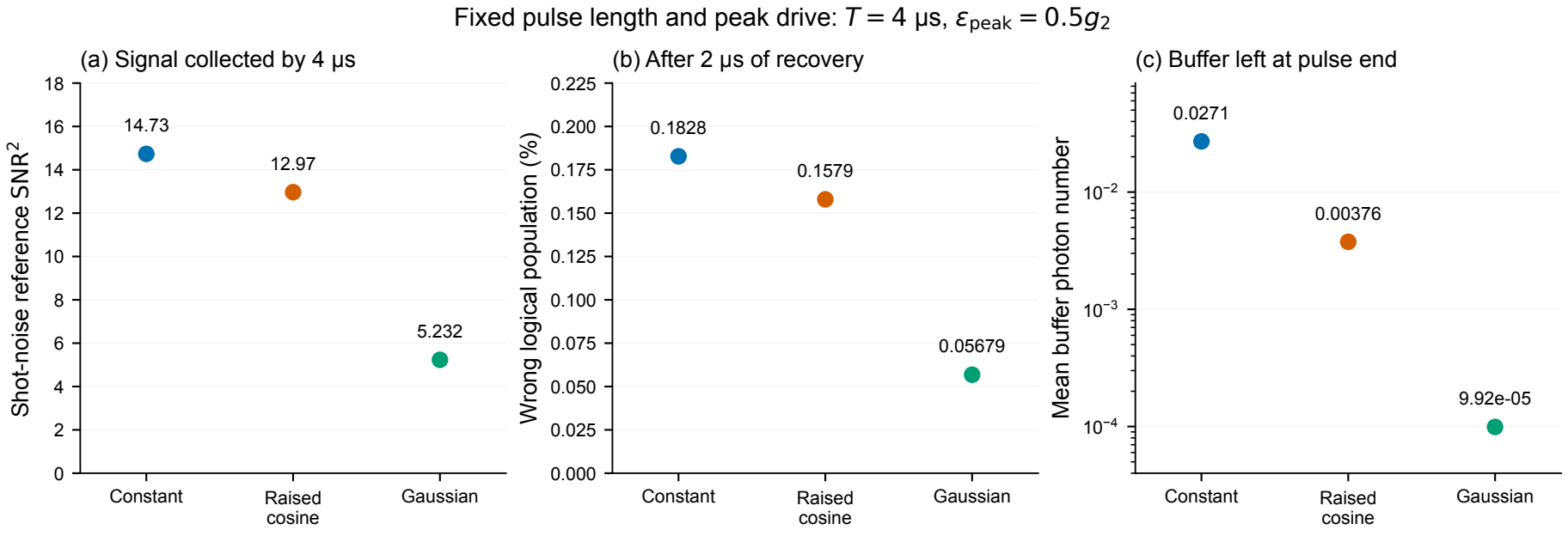
Fixed pulse length and peak drive: $T = 4$ μs, $\varepsilon_{\text{peak}} = 0.5g_2$
(a) Signal collected by 4 μs
Shot-noise reference SNR^2
18
16
14
12
10
8
6
4
2
0
14.73
12.97
5.232
Constant
Raised cosine
Gaussian
(b) After 2 μs of recovery
Wrong logical population (%)
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.1828
0.1579
0.05679
Constant
Raised cosine
Gaussian
(c) Buffer left at pulse end
Mean buffer photon number
10^{-2}
10^{-3}
10^{-4}
0.0271
0.00376
9.92e-05
Constant
Raised cosine
Gaussian